

ANKARA ETIMESGUT, Turkey

WMO: 171290

Lat: 39.950N

Lon: 32.689E

Elev: 2653

StdP: 13.34

Time Zone: 3.00 (E03)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	14.2	18.1	8.2	9.2	20.0	11.8	11.1	23.8	21.0	45.9	18.1	43.9	1.9	280	0.483

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	26.4	95.0	63.8	91.7	63.5	89.3	63.0	68.0	85.7	66.4	83.8	65.1	82.7	7.8	270

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
62.3	92.7	74.2	60.5	86.8	73.3	58.8	81.7	72.5	33.7	86.0	32.5	83.8	31.4	82.4	76.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%		2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
Min		Max		Min	Max		Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)		(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
18.2	15.5	13.3		9.0	99.3	5.3	2.9	5.2	101.4	2.1	103.1	-0.9	104.7	-4.8	106.9	
				7.3	70.0	5.7	2.7	3.2	71.9	-0.2	73.5	-3.4	75.0	-7.6	77.0	
				DB												
				WB												

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	54.2	34.1	36.9	43.5	51.8	61.1	68.3	74.7	75.1	66.1	56.0	44.1	36.8	(6)
	DBStd	15.73	7.80	7.56	7.15	6.51	5.82	5.04	4.60	4.59	5.89	6.73	6.75	7.12	(7)
	HDD50	1766	493	368	220	53	1	0	0	0	0	25	196	410	(8)
	HDD65	4822	958	786	665	398	148	24	1	2	57	284	626	873	(9)
	CDD50	3286	1	2	20	106	346	550	767	778	483	211	20	2	(10)
	CDD65	866	0	0	0	1	28	124	303	315	90	5	0	0	(11)
	CDH74	10961	0	0	3	52	490	1552	3704	3702	1285	173	0	0	(12)
	CDH80	4598	0	0	0	3	110	542	1732	1729	452	30	0	0	(13)
(14) Wind	WSAvg	4.7	3.9	4.5	5.5	5.2	5.1	5.5	6.2	5.8	4.7	3.8	3.0	3.6	(14)
(15) Precipitation	PrecAvg	14.8	1.7	1.3	1.3	1.7	1.9	1.3	0.6	0.5	0.7	0.9	1.1	1.9	(15)
	PrecMax	24.6	3.9	3.5	3.3	3.7	3.5	3.6	2.0	1.7	3.7	2.7	4.6	3.8	(16)
	PrecMin	9.8	0.2	0.3	0.3	0.2	0.7	0.1	0.0	0.0	0.0	0.1	0.1	0.2	(17)
	PrecStd	3.1	1.1	0.9	0.8	1.0	0.7	1.1	0.6	0.4	0.8	0.7	0.9	0.9	(18)
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	55.6	62.7	71.5	79.9	87.4	93.1	99.8	98.6	93.3	84.1	69.8	57.6	(19)
		MCWB	46.1	48.7	50.2	56.7	61.3	63.9	63.9	63.8	62.2	58.3	53.0	49.0	(20)
	2%	DB	51.5	57.2	65.9	74.9	82.7	89.3	95.3	95.0	89.1	78.7	64.8	53.5	(21)
		MCWB	44.8	46.4	49.0	54.9	60.5	62.9	64.4	63.4	61.2	56.6	51.2	46.7	(22)
	5%	DB	48.1	53.5	61.4	70.1	79.6	86.0	91.7	91.8	85.6	74.8	60.8	50.2	(23)
		MCWB	42.8	44.4	47.3	53.1	59.3	62.1	63.8	63.4	60.2	56.1	49.1	45.1	(24)
	10%	DB	44.8	49.7	57.5	66.3	75.5	82.7	89.2	89.3	81.9	70.1	57.0	47.8	(25)
		MCWB	41.1	42.8	46.1	51.5	57.4	60.9	63.5	63.1	59.1	54.6	47.8	43.7	(26)
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	47.6	50.9	52.8	59.3	66.0	68.0	71.9	69.6	66.8	61.0	55.5	50.3	(27)
		MCDB	53.7	59.3	67.0	72.9	82.1	85.5	89.9	87.8	84.4	73.2	65.7	56.0	(28)
	2%	WB	45.1	47.3	50.2	56.3	62.5	65.6	68.3	67.8	64.1	58.9	52.8	47.7	(29)
		MCDB	50.3	55.1	62.0	71.3	78.6	82.6	86.1	85.5	81.2	71.4	62.0	52.2	(30)
	5%	WB	43.1	45.2	48.4	54.2	60.4	63.9	66.4	66.3	62.3	57.6	50.7	45.9	(31)
		MCDB	46.9	52.1	59.2	67.6	75.3	80.0	84.2	83.5	79.1	70.6	59.0	49.8	(32)
	10%	WB	41.2	43.0	46.9	52.4	58.8	62.6	65.0	65.0	60.7	56.1	48.5	43.9	(33)
		MCDB	44.4	48.7	56.5	64.5	72.8	78.2	83.2	82.6	77.3	67.9	55.6	47.3	(34)
(35) Mean Daily Temperature Range	5% DB	MDBR	14.0	18.1	21.0	22.7	24.2	25.1	26.4	26.2	27.0	24.2	21.7	14.3	(35)
		MCDBR	19.2	25.2	29.1	30.2	30.0	30.4	30.5	30.1	33.0	31.6	28.8	18.5	(36)
		MCWBR	13.1	15.4	15.5	14.3	12.4	10.8	9.9	9.7	12.7	14.2	16.1	12.3	(37)
	5% WB	MCDBR	17.0	21.9	24.8	26.8	26.6	26.5	26.6	25.9	27.5	25.5	24.3	16.3	(38)
		MCWBR	12.1	13.9	13.7	13.7	11.8	10.4	10.0	9.4	11.4	12.2	14.1	11.2	(39)
(40) Clear-Sky Solar Irradiance	taub		0.307	0.333	0.376	0.413	0.392	0.372	0.370	0.368	0.348	0.348	0.318	0.307	(40)
	taud		2.380	2.326	2.223	2.148	2.274	2.349	2.354	2.337	2.390	2.387	2.446	2.436	(41)
	Ebn at Noon		271	279	277	273	281	286	286	284	283	270	266	262	(42)
	Edh at Noon		27	33	41	47	42	39	39	38	34	30	25	24	(43)
(44) All-Sky Solar Radiation	RadAvg		595	914	1278	1638	1949	2215	2348	2091	1647	1125	778	529	(44)
	RadStd		73	132	114	146	140	186	99	85	112	81	77	68	(45)

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (10 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page